Venture Financing:
Bootstrapping & Startup Capital

EAS5450 Class #12

Penn Engineering

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Artificial Intelligence (AI) Policy:
We acknowledge the growing prevalence of AI in today's academic and professional environment and the potential it holds for facilitating various tasks. In this course, the use of AI tools is permitted for essay assignments, but not for quizzes, problem sets, or the final exam. AI tools can serve as supplementary aids for research and content organization; however, it is essential that the final output reflects your own understanding, interpretation, and synthesis of the material. AI should be used as an enhancement to your learning experience, not as a substitute for your own analytical and critical thinking. Also, your essay must reflect the content of the assigned reading or case study, and should not include information or data extraneous to, or outside the scope of, the reading or case study. It is important to exercise caution when using AI, particularly for case studies; these readings may contain historical or modified information that is not available or verified through internet sources. In addition, AI tools can sometimes generate misleading or inaccurate 'hallucinations'—information that appears factual but is not. Therefore, always cross-verify the information provided by AI tools to ensure its accuracy and relevance to the assignment.

In every instance where AI is used, you are required to disclose its use and appropriately cite all sources, including AI-generated content or suggestions. Failure to do so will be considered a violation of the university Code of Academic Integrity. The use of AI should augment your learning experience, not replace your own intellectual work. Your engagement with the course content is crucial in fostering a rich learning environment and developing critical skills in technology innovation, marketing, and product management.

NOTE: This AI policy is an edited version of one drafted using ChatGPT 4.0 from OpenAI where the course syllabus was input and ChatGPT was asked to draft an AI policy for the course that allows its use for written essay assignments, excluding quizzes, on readings or case studies.

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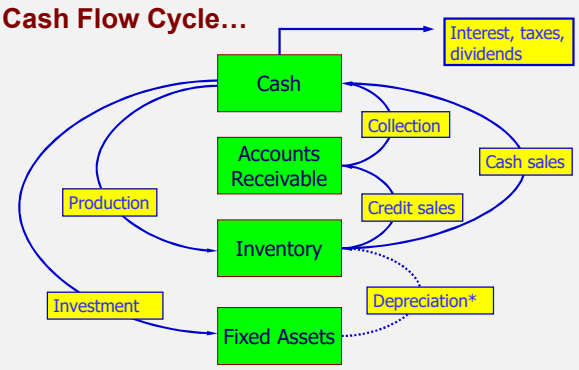
Recall…

Cash Flow Cycle…



* Non-Cash Item

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"Cash is King"

… not net income, profits or earnings.

Peter Drucker (1909-2005)

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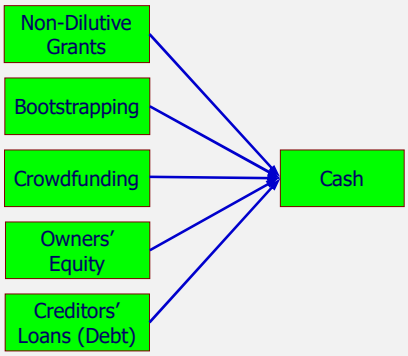
Priming the Cash Flow Cycle



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Equity and Debt …

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- **EQUITY**
  - Buys ownership shares
  - Voting rights
  - ROI = dividends + value appreciation
  - Dividends = discretionary f(earnings)
  - Subject to performance risk
  - Shares in the upside
- **DEBT**
  - Loan: must be repaid with interest
  - Specified duration
  - Senior obligation
  - Default rights
  - Typically secured by assets
  - No ownership rights
  - No share in upside

Pure Equity (Common Stock) — *Continuum of Securities* — **Pure Debt (Term Loan)**

Common Stock | Restricted Common Stock | Preferred Stock | Convertible Preferred Stock | Convertible Debt | Term Loan

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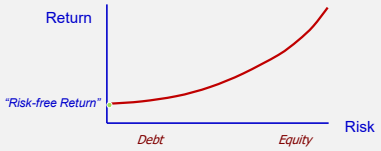
Equity is more expensive than debt…

- Return (cost of capital) is commensurate with risk.
- Debt is a senior obligation (i.e., it gets repaid first) and is, therefore, less risky.
- Equity buys ownership in the firm, and shares ownership risk (i.e., is subject to performance risk).



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… But Start-Ups Are Not "Bankable"

- High risk perception by banks
- No track record
- Lack of collateral assets (property, plant, equipment)
- No accounts receivable

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Accessing Start-Up Debt

- Personal loan guarantees
- Home mortgage-backed loans
- Gov't loan guarantees and grants
- *Eventually:* balance sheet financing
  - Asset-backed loans
    - Property, plant and equipment
    - Accounts receivable
  - Track record

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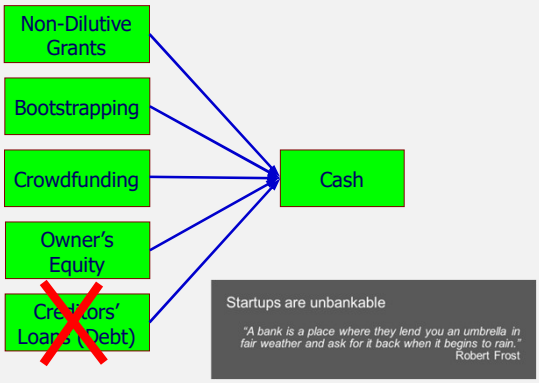
Priming the Cash Flow Cycle



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Bootstrap: Just Do It

- Personal founders' funds launched **80%** of Inc. "500" companies
- Median start-up capital = $10,000
- Follow-on growth financing:
  - **<20%** raise outside equity
  - Most rely on debt and retained earnings

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Bootstrap Finance: The Art of Start-Ups

Amar Bhide

- "Raising money has become a disease…"
  - Distraction from the firm's mission
  - Hinders flexibility (things go wrong!)
  - Spending doesn't fix problems
  - Loss of ownership
  - Loss of control
  - The *Spoiled Startup*

The Golden Rule:

He or she who has the gold… rules!

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- **Seven Axioms of Successful Entrepreneurs**
  1. Get operational quickly
     … opportunities develop when you're on the playing field.
  2. Look for quick cash flow (e.g., consulting)
  3. Offer high value products/services
     … direct selling by the entrepreneur is an advantage.
  4. Forget about the crack team
     … you can't afford it.
  5. Keep growth in check
     … expand at a rate you can afford and can control.
  6. Focus on cash, not profits
  7. Cultivate banks
     … before you're credit worthy.

Bhide, A. "Bootstrap Finance: The Art of Start-Ups"

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But, if you have no bootstrapping options…

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Sources of Seed Capital

- Personal savings: 74%
- Private investors ("angels"): 7%
- Strategic corporate investment: 6%
- Family and friends: 5%
- Venture capital funds: 5%
- Public stock offerings: 3%
- Initial coin offerings (ICOs): ?

… and Crowd-Funding

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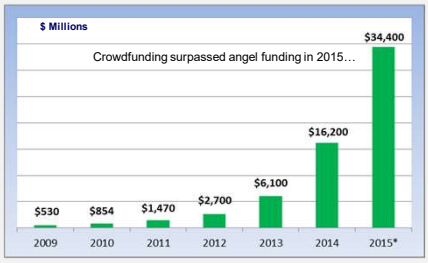
Crowd-Funding is often a non-dilutive way to fund (and test) a product

Enacted 2012



Source: Massolution: Crowdsourcing.org

* 2015 Estimated

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Popular Crowd-Funding platforms…

Site	Total Raised	Supporters	Platform Fee	Payment Fee	Important to Know
gofundme	$5B	50M	0%	2.9% + $0.30	✔ Can withdraw immediately and deposits take 2-5 business days ✔ 24/7 rapid email support, mobile app, superior add beneficiary feature ✔ Easy to use fundraising tools make setup fast ✔ GoFundMe Guarantee protects donors and beneficiaries from fraud
INDIEGOGO	$1.5B	10M	5%	3.0% + $0.30	✔ Offers "flexible funding" ✔ Specializes in technology and hardware product launches ✔ Regular email support hours; marketing and campaign strategy support
KICKSTARTER	$3B	15M	5%	3.0% + $0.20	✔ Specializes in creative projects with robust reward level feature ✖ 14-day wait to withdraw and deposits take 5-7 business days ✖ Limited email support hours ✖ Requires Kickstarter approval to launch a fundraiser
FUNDLY	$330M	NA	4.9%	2.9% + $0.30	✔ Can withdraw immediately and deposits take 2-5 business days ✖ No donor guarantee policy for fraud protection ✖ Limited email support hours

Crowdfunding.com (2017)

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KICKSTARTER

Coffee Joulies:
- May 2, 2011
- Sought $9,500
- 4,818 backers
- $307,000 raised
- No shares offered
- No loss of control

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Pebble Watch:
- May 2012
- Sought $100,000
- 68,929 backers
- $10,266,845 pledged
- No shares offered
- No loss of control

Coolest Cooler:
- Aug 2014
- Sought $50,000
- 62,642 backers
- $13,285,226 pledged
- No shares offered
- No loss of control

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- Private Equity?
- Venture Capital?
- Hedge Fund?

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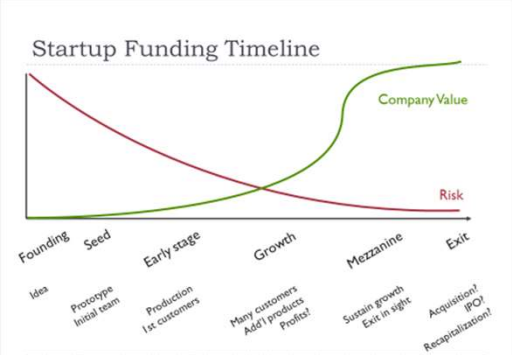

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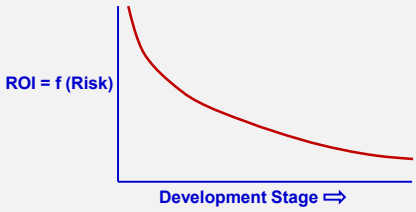
Equity Financing Stages

- Early stage
- Expansion
- Exit



ROI: 'Return on Investment' = cost of capital

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Higher Risk = Higher Return

Type of Fund	Risk	20-Yr ROI % p.a.
Early/Seed VC	High	26.7 %
Later/Expansion Stage VC	Moderate	16.5 %
Balanced VC	Moderate	14.1%
NASDAQ	Low*	8.5 %
S&P 500	Low*	8.6 %

* NASDAQ and S&P equity market risk is low relative to venture capital risk, but is nevertheless risky.

Source: Thomson ONE – U.S. Private Equity Performance Index as of March 31, 2012
Cambridge Assocs LLC – U.S. Venture Capital Index, the performance index of the National Venture Capital Assn (NVCA), Q1 2012

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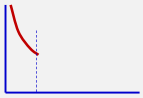
Early-Stage Financing



- **Founders' Money**
  - Proof of concept / Prototype development
- **Seed Financing**
  - Product development / Initial marketing
- **First-stage financing (Series A)**
  - Initiate manufacturing and sales
  - Unlikely to be profitable, yet

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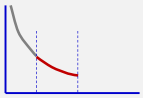
Expansion Financing



- **Second-stage (Series B)**
  - Producing and shipping / Product improvement
  - May not yet be profitable
- **Third-stage (Series C)**
  - Cash to sustain growth
  - Breaking even or profitable

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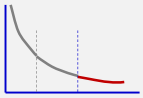
Mezzanine and Exit



- **Bridge ("Mezzanine")**
  - More cash to sustain growth
  - VC cash out is foreseeable
- **Exit**
  - IPO: Public markets buy stock.
  - Strategic Investor: Acquiring company buys stock.
  - Recapitalization: Company/management buys back stock with new debt and/or equity.

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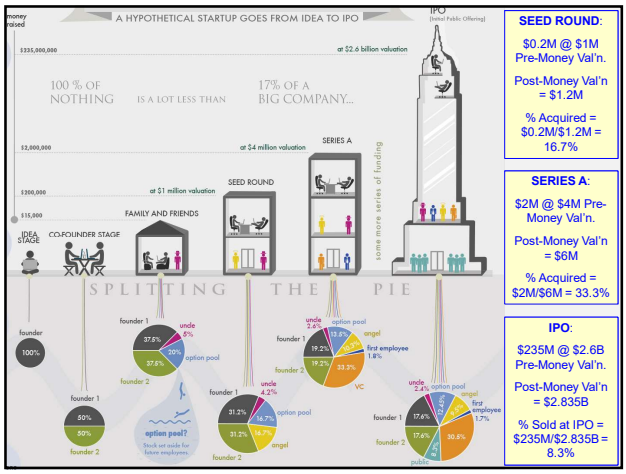

SEED ROUND:

$0.2M @ $1M Pre-Money Val'n.

Post-Money Val'n = $1.2M

% Acquired = $0.2M/$1.2M = 16.7%

SERIES A:

$2M @ $4M Pre-Money Val'n.

Post-Money Val'n = $6M

% Acquired = $2M/$6M = 33.3%

IPO:

$235M @ $2.6B Pre-Money Val'n.

Post-Money Val'n = $2.835B

% Sold at IPO = $235M/$2.835B = 8.3%

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Authorized & Issued Stock Shares

- **Shares authorized at time of incorporation**
  - Fixed maximum number of shares authorized for future issuance
  - Specified in company's Certificate of Incorporation
  - Held by company until issued
  - Par value of shares: $1/Shr typ.
- **Shares issued to shareholders after incorporation**
  - Drawn from authorized shares and sold to shareholders
  - Shares representing company ownership by shareholders
  - Possessed by the shareholders
  - Purchase price received by company treasury upon issuance

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High-Tech Investors Look for…

1. **Visionary leader & capable team**
   - "We can hire execution"
2. **Market leadership**
   - Emerging growth markets with sizable potential
   - Some paying customers now
3. **Fundamental, compelling technology advantage**
   - Defensible competitive position
   - Proprietary IP
4. **Capital-efficient business plan**
   - Bootstrapped prototype
   - Moderate capital requirements
   - Clear return on investment
   - Easily financed post-seed
   - Fast path to profitability
5. **Risks mitigated in incubation stage**

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"It's all about managing risk"

1. Technical… *Does it work?*
2. **Market…** ***Enough customers?***
3. Operating… *Can it be built profitably?*
4. Distribution… *Cost-effective channel?*
5. Team… *Can they execute?*

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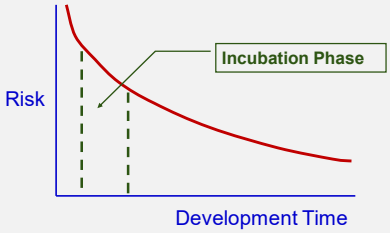
Incubation

- Remove as much risk for as little money in as short a time as possible.



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Incubation Transformation

- "PROJECT"
  - Idea
  - Exciting technology
  - Capable founders
- "COMPANY"
  - Proven technology
  - Demonstrated prototype
  - Market position
  - Viable business plan
  - Management team
  - Financing

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Accelerators

Y Combinator — Mountain View

500 startups — Mountain View

techstars — Boulder, NYC, Boston, Seattle

Dreamit Ventures — Philadelphia, NYC, Tel Aviv, Austin

The standard deal
- 3 month bootcamp
- Stipend
- Mentorship/Connections
- Curriculum
- Street cred
- Free legal, accounting
- Demo day with investors
- 6% stake (convertible debt)

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Investors Must Foresee an Exit Strategy

- **IPO**: Underwriters sell the company's stock in the public market.
- **Strategic Investor**: Acquiring firm buys the company's stock.
- **Recapitalization**: Company buys back its investors' stock with new debt and/or equity.

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Private Equity is Cyclical

- **Private equity markets = f (public stock markets)**
- **Low public markets drain money from private equity:**
  - Attractive alternative investment bargains
  - Lower investor net worth
  - Poor liquidity (IPOs, Mergers, Acquisitions)
    - Can't foresee an "exit"

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In a down (stock) market…

- **VCs fund few new deals -- preserve $ for existing portfolio**
- **VCs cannot rely on subsequent rounds:**
  - Line up future funding needs upfront
  - Milestone reviews
  - Little flexibility and patience
  - Low valuations
    - Invest less and take more

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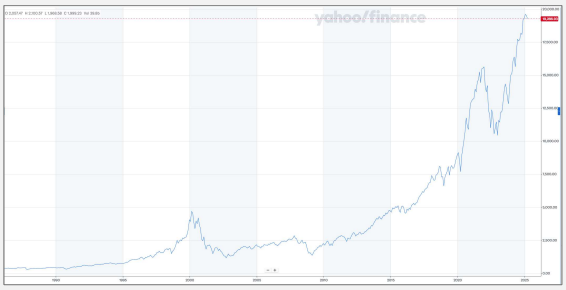
NASDAQ Performance



Yahoo Finance: Feb 24, 2025

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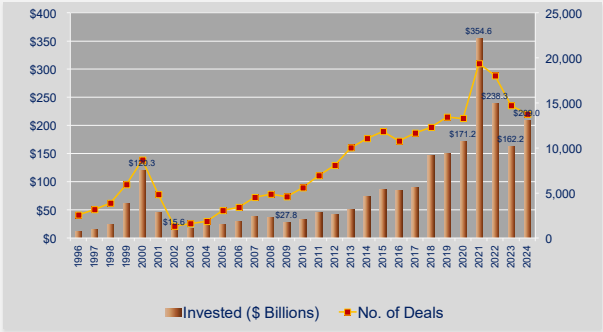
U.S. Venture Capital



Source: PitchBook NVCA Venture Monitor, 2022

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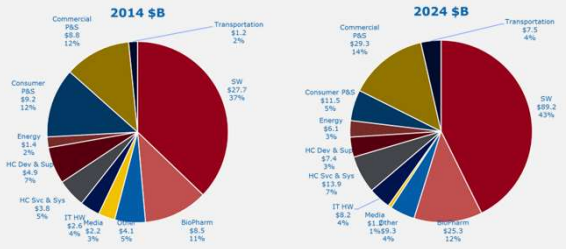
U.S. VC Investment by Sector



Source: PitchBook NVCA Venture Monitor, 4Q2024

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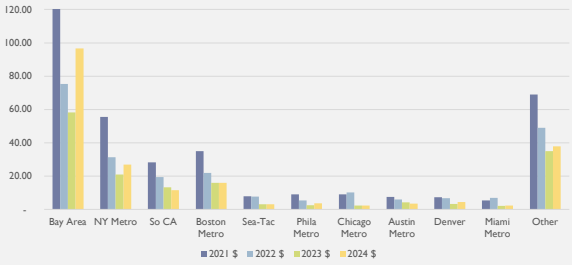
VC Investment Geographic Distribution
($ Billions)



Source: PitchBook NVCA Venture Monitor, 2022

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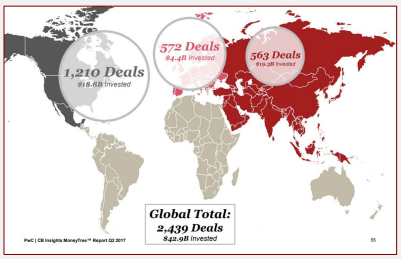
2017Q2, for the 1st time ever, Asian Startups had more Invested than North America



For every dollar invested globally, 45 cents went to Asian firms, including Didi Chuxing, which raised a massive $5.5B round, the largest round ever raised by a VC-backed company.

— ***Suneet Dua**, Partner, Network Leader at PwC*

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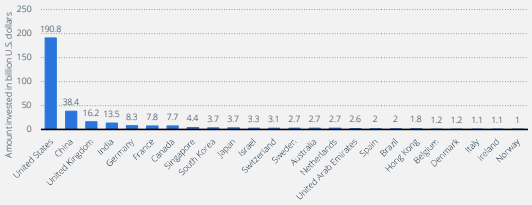
Leading Countries for VC Investment in 2024



Note(s): Worldwide
Further information regarding this statistic can be found on page 6.
Source(s): Dealroom.co; ID 1466489

statista

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China's Private Sector has been Losing Ground to State-Owned Enterprises in Recent Years

SOURCE: Financial Times, 2024.

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Angel Entities

Individual Angels

- Typically entrepreneurs who have cashed out
- Contribute:
  - Patient capital
  - Knowledge and experience
  - Mentoring

Angel Groups/Clubs

- Same as Individual Angels, plus:
- Process to recruit, "scrub" and present opportunities to Angels

Angel Funds

- Same as Angel Groups, plus
- Formal fund, usually set up as LLCs
- Invest as a fund, although Angels direct the investment strategy

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Angel Characteristics

- **Invest close to home**
- **Harvested entrepreneurs**
  - Value-added
    - Experience
    - Network contacts
    - Credibility
  - Patient Capital
- **Regional Networks**

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30X more angels invested 11% of funds in 4X more deals than VC's in 2023

Angels[1]
- 422,350 active angels
- Invested $18.6B in 54,375 ventures
- Avg $340K per deal
- Earlier stage (typ.)
- 1 in 4 pitches funded

Venture Capital[2]
- 474 active V.C. funds
- $170.6B invested in 13,608 deals
- Avg $13M per deal
- Later stage (typ.)

Center for Venture Research, 2023

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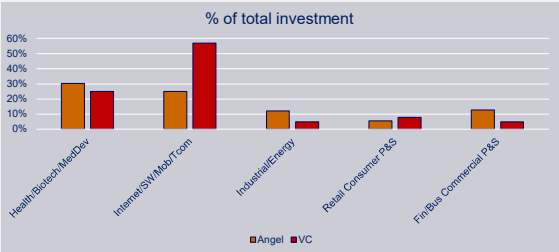
Angel vs VC Sector Investment - 2023



Source: Center for Venture Research 2024
Pitchbook NVCA Venture Monitor 2024

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Online Angel – Startup Meeting

AngelList

FundersClub

gust

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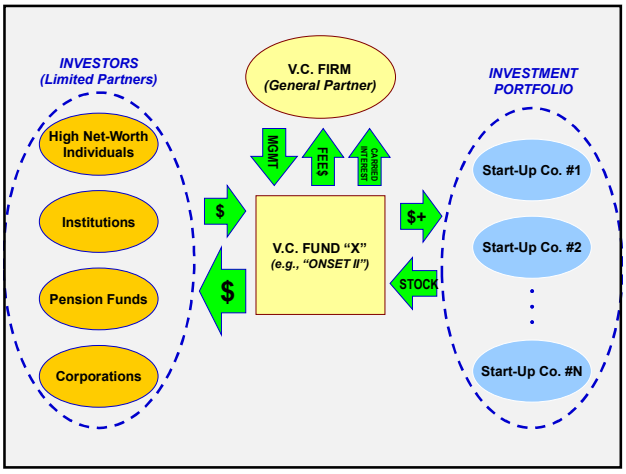

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While the choice is yours, choose investors who can (and <u>will</u>) provide:

- Experienced mentoring
- Sophistication, understanding
- Staying power and patience
- Future cash infusions
- Credibility
  - Access to future investors
  - Access to strategic partners

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- **Be sure you can work with them…**
  And that they will work with you.

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6 Principles of Successful Startups

ONSET VENTURES

1. Experienced mentor.
2. Flexibility with business model.
3. Defer CEO hiring until model validated.
4. Create value for next investment round.
5. There must be a *want* for the product.
6. People: missionaries not mercenaries
   - Solid understanding of technology
   - Ability to communicate articulately

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★ Bootstrap: Just Do It ★

- **Personal founders' funds launched 80% of Inc. "500" companies**
  - Avoids distraction from mission
  - Enhances flexibility
  - Maintains founder ownership
  - Maintains founder control
  - Doesn't "spend over" problems
  - Controlled growth

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